

Ext 1



NORTH SYDNEY BOYS HIGH SCHOOL

2017 Year 11 Task 2

Mathematics Extension 1

General Instructions

- Working time – 50 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	Total	Total %
Mark	$\frac{7}{7}$	$\frac{7}{7}$	$\frac{7}{7}$	$\frac{6}{6}$	$\frac{4}{4}$	$\frac{6}{6}$	$\frac{5}{5}$	$\frac{42}{42}$	$\frac{100}{100}$

Question 1

Differentiate with respect to x .

(a) $5x^4 - x$ 1

(b) $\frac{1}{x^2 + 2}$ 1

(c) $2(5 - 2x^2)^{-3}$ 1

(d) $(2x + 1)(3x - 2)^4$ 2

(e) $\frac{\sqrt[3]{x}}{x}$ 2

Question 2

(a) Find the acute angle between the two lines with equations $y = 4x - 4$ and $3x - 6y + 1 = 0$ 2

(b) Find the coordinates of the point P, that divides externally the interval AB in the ratio 3 : 4 where $A(-5, 7)$ and $B(9, 0)$. 2

(c) The distance between the point $P(1 + k, 2 - k)$ and the line $3x + 4y - 1 = 0$ is 5 units. Find the possible values of k . 3

examination continues overleaf.....→

Question 3

- (a) Prove that the point (2,3) lies on the line whose equation is
 $x - y + 1 + k(2x + y - 7) = 0$ 1
- (b) Sketch the region which represents the **intersection** of
 $y > x^2 - 4$ and $y \leq 2x - 1$ showing points of intersection. 3
- (c) Sketch the region $x < -\sqrt{9 - y^2}$ 3

Question 4

- (a) Find a simple expression for
 $\cos 2x \cos 3x - \sin 2x \sin 3x$ 1
- (b) Solve $\cos \frac{\theta}{2} = \frac{1}{2}$ for $-\pi \leq \theta \leq \pi$ 2
- (c) Solve $\sin 2\theta = \tan \theta$ for $0^\circ \leq \theta \leq 360^\circ$ 3

Question 5

- (a) Prove $\frac{\sin 4A + \sin 2A}{1 + \cos 4A + \cos 2A} = \tan 2A$ 2
- (b) Find the possible values of $\sin \frac{A}{2}$ if $\cos A = \frac{3}{5}$. 2

examination continues overleaf.....→

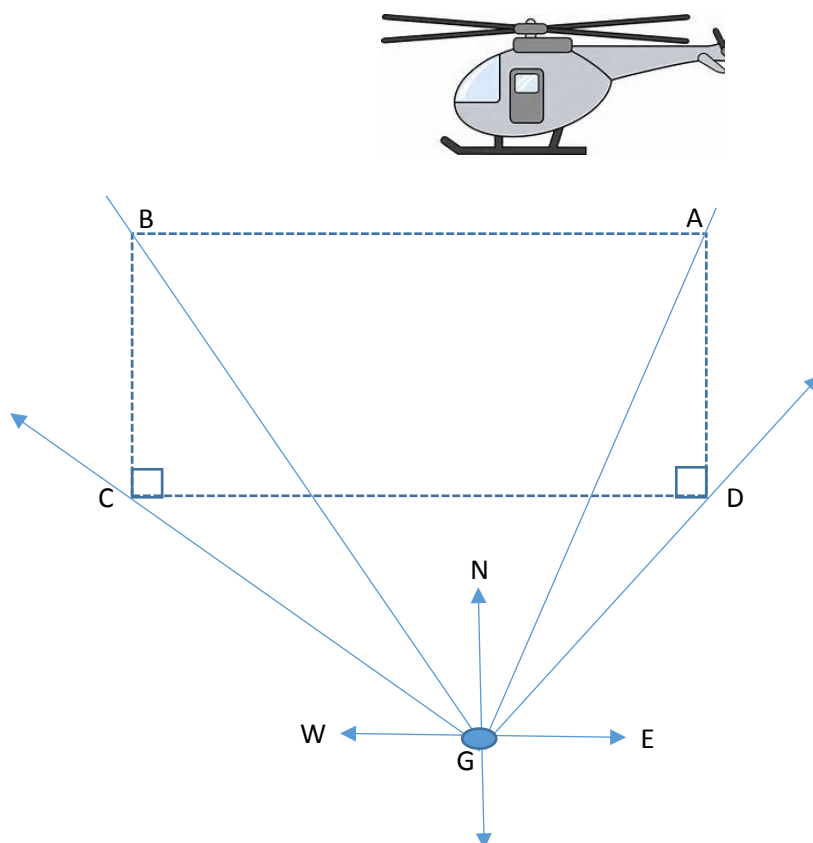
Question 6

- (a) For what values of x is the gradient of the curve $y = x - \frac{4}{x}$ greater than 5? 3
- (b) If the tangent to the curve $y = \frac{ax+b}{(x-1)(x-4)}$ at the point $(2, -1)$ is parallel to the x -axis, find the values of a and b . 3

Question 7

A helicopter flies due west from A to B at a constant speed of 420km/h. From a point G on the ground the bearing of the helicopter when it is at A is $079^\circ T$ with an angle of elevation β . Four minutes later the helicopter is at B with a bearing from G being $302^\circ T$ and an angle of elevation 32° . The altitude of the helicopter is h metres.

- (a) Copy and complete the sketch below showing the above information. 1
- (b) Calculate the height of the plane to the nearest metre. 2
- (c) Calculate the value of β to the nearest degree. 2



END OF EXAMINATION

Question 1

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① a) $5x^4 - x$

$$\frac{dy}{dx} = 20x^3 - 1$$

b) $(x^2 + 2)^{-1}$

$$\frac{dy}{dx} = -1(x^2 + 2)^{-2} \cdot 2x$$

$$= \frac{-2x}{(x^2 + 2)^2}$$

c) $2(5 - 2x^2)^{-3}$

$$\frac{dy}{dx} = -6(-4x)(5 - 2x^2)^{-4}$$

$$= 24x(5 - 2x^2)^{-4}$$

d) $(2x+1)(3x-2)^4$

$$\frac{dy}{dx} = 2(3x-2)^4 + (2x+1) \cdot 4 \cdot 3(3x-2)^3$$

$$= 2(3x-2)^4 + 12(2x+1)(3x-2)^3$$

e) $\frac{\sqrt[3]{x}}{x} = \frac{x^{\frac{1}{3}}}{x}$

$$= x^{-2/3}$$

$$\frac{dy}{dx} = \frac{-2}{3} x^{-5/3}$$

$$= \frac{-2}{3 x^{5/3}}$$

$$= \frac{-2}{3 \sqrt[3]{x^5}}$$

Question 2

$$(a) m_1 = 4$$

$$m_2 = \frac{1}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{4 - \frac{1}{2}}{1 + 2} \right|$$

$$= \left| \frac{\frac{7}{2}}{3} \right|$$

$$= \frac{7}{6}$$

$$\theta = 49^\circ 24'$$

$$b) \begin{matrix} m & n \\ -3 & 4 \end{matrix} \quad \begin{matrix} x_1, y_1 \\ (-5, 7) \end{matrix} \quad \begin{matrix} x_2, y_2 \\ (9, 0) \end{matrix}$$

$$(x, y) = \left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n} \right)$$

$$= \left(\frac{4(-5) - 3(9)}{-3+4}, \frac{4(7) - 3(0)}{-3+4} \right)$$

$$= \left(\frac{-47}{1}, \frac{28}{1} \right)$$

$$= (-47, 28)$$

$$c) d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$5 = \frac{|3(1+k) + 4(2-k) - 1|}{\sqrt{9+16}}$$

$$25 = |3 + 3k + 8 - 4k - 1|$$

$$25 = |10 - k|$$

$$10 - k = 25 \quad \text{or} \quad 10 - k = -25$$

$$k = -15 \quad \text{or} \quad k = 35$$

Question 3

a) Sub (2,3)

$$\text{LHS} = x - y + 1 + k(2x + y - 7)$$

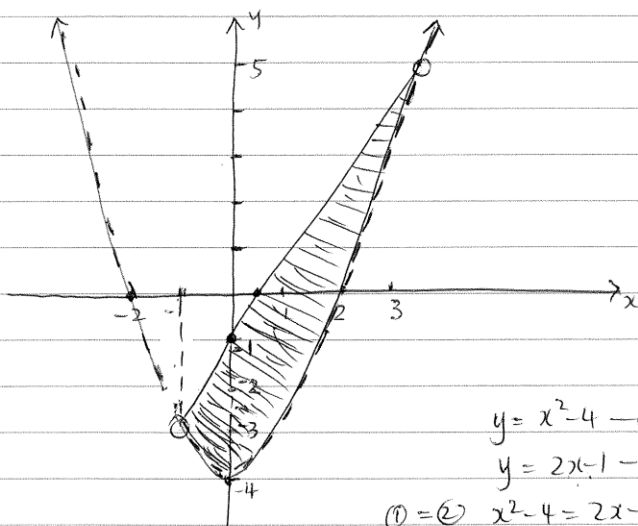
$$= 2 - 3 + 1 + k(4 + 3 - 7)$$

$$= 0$$

$$= \text{RHS}$$

$\therefore (2,3)$ lies on line above.

b) $y > x^2 - 4$ and $y \leq 2x - 1$



$$y = x^2 - 4 \quad \text{--- (1)}$$

$$y = 2x - 1 \quad \text{--- (2)}$$

$$\text{(1) = (2)} \quad x^2 - 4 = 2x - 1$$

$$x^2 - 2x - 3 = 0$$

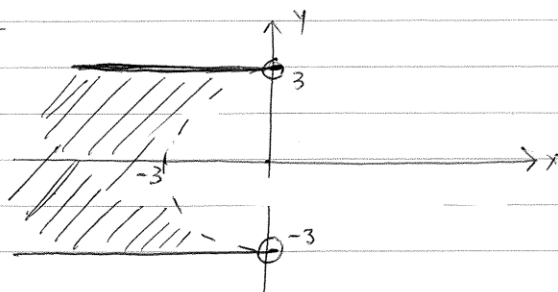
$$(x - 3)(x + 1) = 0$$

$$x = 3, y = 5$$

$$x = -1, y = -3$$

c) $x < -\sqrt{9 - y^2}$

$$-4 < -\sqrt{9 - 0}$$



Question 4

$$\begin{aligned}\text{(a)} \quad & \cos 2x \cos 3x - \sin 2x \sin 3x \\ &= \cos(2x+3x) \\ &= \cos 5x\end{aligned}$$

$$\begin{aligned}\text{b)} \quad & \cos \frac{\theta}{2} = \frac{1}{2} & -\pi \leq \theta \leq \pi \\ & & -\frac{\pi}{2} \leq \frac{\theta}{2} \leq \frac{\pi}{2} \\ & \frac{\theta}{2} = \frac{\pi}{3}, -\frac{\pi}{3}.\end{aligned}$$

$$\theta = \frac{2\pi}{3}, -\frac{2\pi}{3}$$

$$\text{c)} \quad \sin 2\theta = \tan \theta \quad 0^\circ \leq \theta \leq 360^\circ.$$

$$2\sin \theta \cos \theta = \frac{\sin \theta}{\cos \theta}$$

$$2\sin \theta \cos^2 \theta - \sin \theta = 0$$

$$\sin \theta (2\cos^2 \theta - 1) = 0$$

$$\sin \theta = 0 \quad \text{or} \quad 2\cos^2 \theta - 1 = 0$$

$$\cos^2 \theta = \frac{1}{2}$$

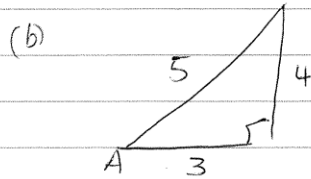
$$\theta = 0, 180^\circ, 360^\circ \quad \cos \theta = \pm \frac{1}{\sqrt{2}}$$

$$\text{or} \quad \theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

Question 5

$$a) \frac{\sin 4A + \sin 2A}{1 + \cos 4A + \cos 2A} = \tan 2A$$

$$\begin{aligned} \text{LHS} &= \frac{\sin 4A + \sin 2A}{1 + \cos 4A + \cos 2A} \\ &= \frac{2\sin 2A \cos 2A + \sin 2A}{1 + 2\cos^2 2A - 1 + \cos 2A} \\ &= \frac{\sin 2A (2\cos 2A + 1)}{2\cos^2 2A + \cos 2A} \\ &= \frac{\sin 2A (2\cos 2A + 1)}{\cos 2A (2\cos 2A + 1)} \\ &= \tan 2A \\ &= \text{RHS.} \end{aligned}$$



$$\cos A = \frac{1 - 2\sin^2 \frac{A}{2}}{2}$$

$$\frac{3}{5} = \frac{1 - 2\sin^2 \frac{A}{2}}{2}$$

$$\frac{3-1}{5} = \frac{-2\sin^2 \frac{A}{2}}{2}$$

$$\frac{1}{5} = \sin^2 \frac{A}{2}$$

$$\sin \frac{A}{2} = \pm \frac{1}{\sqrt{5}}$$

Question 6

$$y = x - 4x^{-1}$$

$$(a) \quad \frac{dy}{dx} = 1 + \frac{4}{x^2}$$

$$1 + \frac{4}{x^2} > 5$$

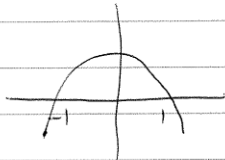
$$\frac{4}{x^2} > 4$$

$$4 > 4x^2$$

$$4 - 4x^2 > 0$$

$$4(1 - x^2) > 0$$

$$4(1-x)(1+x) > 0$$



$$-1 < x < 1 \quad x \neq 0$$

$$(b) \quad y = \frac{ax+b}{(x-1)(x-4)}$$

$$y = \frac{ax+b}{x^2-5x+4}$$

$$\frac{dy}{dx} = \frac{(x^2-5x+4) \cdot a - (ax+b) \cdot (2x-5)}{(x^2-5x+4)^2}$$

When $x=2$, $m=0$

$$0 = (4-10+4)a - (2a+b)(-1)$$

$$0 = -2a + 2a - b$$

$$\therefore b = 0$$

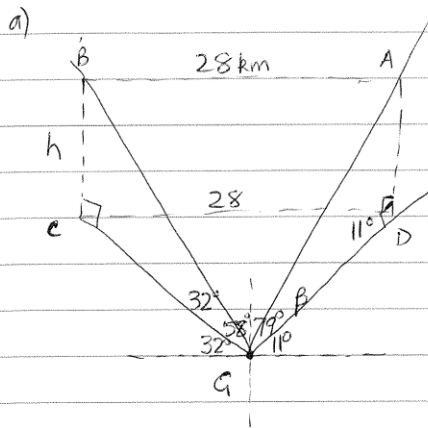
Sub $(2, -1)$ into y and $b=0$.

$$-1 = \frac{2a}{-2}$$

$$a = 1$$

$$\therefore a=1, b=0$$

Question 7



b) $1h = 420 \text{ km}$

$$t_{\min} = \frac{420 \times 4 \text{ km}}{60} = 28 \text{ km}$$

Find CG

$$\frac{CG}{\sin 11^\circ} = \frac{28}{\sin (58+79)}$$

$$CG = \frac{28 \sin 11^\circ}{\sin 137^\circ}$$

Find h

$$\tan 32^\circ = \frac{h}{CG}$$

$$h = CG \tan 32^\circ$$

$$h = \frac{28 \sin 11^\circ \tan 32^\circ}{\sin 137^\circ}$$

$$h = 4895 \text{ m}$$

Find GD

$$\frac{GD}{\sin 32^\circ} = \frac{28}{\sin 137^\circ}$$

$$GD = \frac{28 \sin 32^\circ}{\sin 137^\circ}$$

$$\tan \beta = \frac{h}{GD}$$

$$\beta = 13^\circ$$